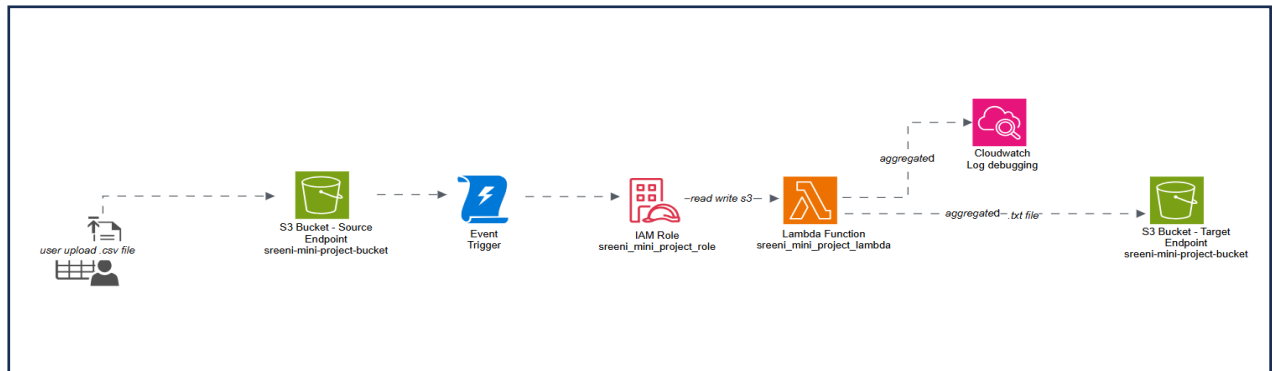


Simple serverless data processing using AWS services - Process Flow Diagram



AWS Services build and use

S3 bucket:	sreeni-mini-project-bucket	Stores input and output files
S3 Event Trigger:	sreeni-mini-project-bucket	Connects upload events to Lambda
IAM role:	sreeni_mini_project_role	Allows Lambda to access S3 and CloudWatch
Lambda function:	sreeni_mini_project_lambda	Processes the uploaded data
CloudWatch:	sreeni_mini_project_lambda	Monitors Lambda execution

End-to-End Flow

- The old project resources were removed.
- New S3 bucket source starting point creation
- IAM role setup
- Lambda function created
- S3 bucket trigger was configured / setup
- Upload the ".CSV file" to [S3 Source Starting Point]
- S3 detects the new object upload
- S3 trigger sends an event to Lambda
- Lambda reads the CSV file from S3 source bucket
- Lambda calculated the salary total by locations
- Lambda writes the output file "aggregated.txt" back to S3 Target endpoint
- The output file can be verified directly from [S3 bucket Target Endpoint]

1. S3 Bucket

1. This is the starting point of the workflow.
2. The user uploads the CSV file to the S3 bucket.
3. The bucket acts as the source location for the Lambda function.

2. Lambda Function

4. Lambda reads the file from S3.
5. It parses the CSV data.
6. It calculates the total salary by location.
7. It writes an aggregated output file back to S3.

3. IAM Role

8. Gives Lambda permission to access AWS resources.
9. Lambda uses the IAM role to read the uploaded file from S3.
10. It also writes the output file back to S3.
11. The role allows Lambda to log events and use basic execution permissions.

4. S3 Trigger

12. Automatically starts the Lambda function when a new file is uploaded.
13. When a .csv file is uploaded to the bucket, S3 sends an event to Lambda.
14. Lambda receives the event and starts processing the file.

5. CloudWatch Logs

15. Stores Lambda execution logs.
16. If the function runs successfully, logs are stored.
17. If there is an error, the logs help identify the issue.